4 **(a)** 1. Waves must meet π rad out of phase. A1

2. Waves must have equal amplitude. A1

(b) **(i)** Wavetrains from S_1 and S_2 are <u>coherent</u> and <u>superpose</u> at points along YZ. C1

When path difference is an integral multiple of λ, the waves meet in phase, <u>constructive interference</u> takes place to give a series of maxima. C1

(ii) It decreased to one quarter of the original x (since $x = \lambda D/a$) A1

(iii) The line YZ is not parallel to the slits A1

or the slits not normal to the (incident) microwaves

(iv) Place a polariser in front of the transmitter and rotate through 90° OR rotate transmitter/detector through 90°. M1

If this causes minimal/zero signal at some angles, the wave is plane polarized. A1

(c) Distance between two nodes = ½ λ = speed of detector / frequency of detection C1

= 10 / 1.5 = 6.7 mm

Hence, wavelength = 13 mm C1

$f = c/\lambda$□$= 3.00 \times 10^{8} / 13 \times 10^{-3} = 2.3 \times 10^{10}$ Hz. A1

(d) **(i)** White light diffracts after passing through the slits in the grating. For zeroth order maxima, each of the wavelengths in the white light travels the same path length/ zero path difference. B1

The amplitudes add up vectorially to produce a resultant white colour maxima A1

(ii) wavelength λ of red light > wavelength blue light ($\lambda_{red} > \lambda_{blue}$) B1

For waves from any two adjacent slits, path difference is $d \sin\theta$, where d is the separation between the slits. A0

To produce a maxima for 1st order, path difference, $d\sin\theta = 1\lambda$

Hence, maxima for different colors occurs at different angle θ , with red light at a larger angle B1